Pleasant Predictions with Climate Wins:

Harnessing Machine Learning to Forecast Weather

Tiffany Kinney August 2024

Introduction

In recent decades, the rise in extreme weather events has highlighted the urgent need for better tools to predict and adapt to climate changes. This presentation, created for ClimateWins, explores how machine learning can provide the advanced forecasting methods necessary to prepare for our evolving climate.

Data Source:
Collected by the European Climate Assessment & Data Set project from 18 different weather stations across Europe from the 1960 to 2022.

Objectives & Overview

Objective: Identify and predict emerging weather patterns and trends in Europe.

Identify unusual weather patterns and changing trends compared to historical data in Europe.

Detect any regional differences and determine which locations may be safest or most pleasant now and in the future.

Monitor for and predict extreme weather events, such as tornadoes, in Europe.

Algorithms

Application

Random Forests
An ensemble learning method that builds multiple decision trees and combines their outputs to improve accuracy and reduce overfitting,

Random Forest can analyze multiple weather factors like temperature and precipitation, identifying unusual patterns and changes by comparing them to historical data.

Convolutional Neural Network (CNN):
A CNN recognizes spatial patterns in images and is well-suited for analyzing large, complex visual data.

CNNs for Regional Weather Prediction
Analyze radar & satellite images to recognize geographic features & specific locations, and associate them with collected weather data, improving prediction accuracy for local weather patterns.

Generative Adversarial Network (GAN)
Artificial intelligence model made up of Two models (generator & discriminator) compete to improve each other to create synthetic data.

Tornado analysis and prediction
Discover hidden patterns in radar data too complex for humans to notice, predict whether a storm cell might generate a tornado, based on the patterns it learned.

Weather Pattern Analysis

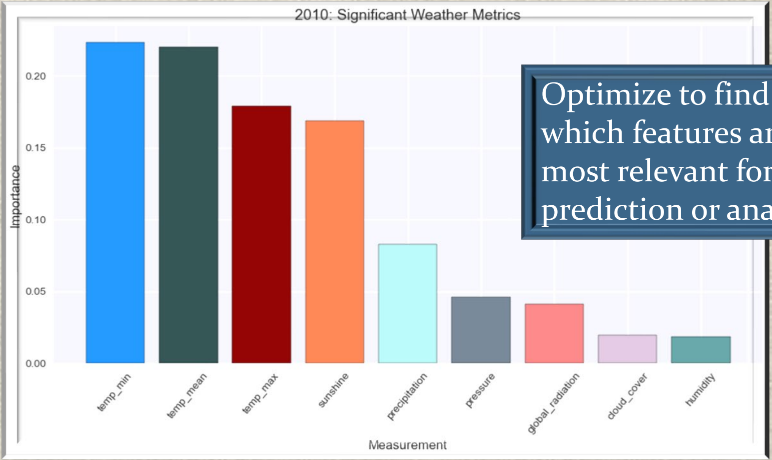
How? Random Forest

- **Multifactor Analysis:** Evaluate multiple weather metrics (precipitation, temperature, wind speed) to predict patterns.

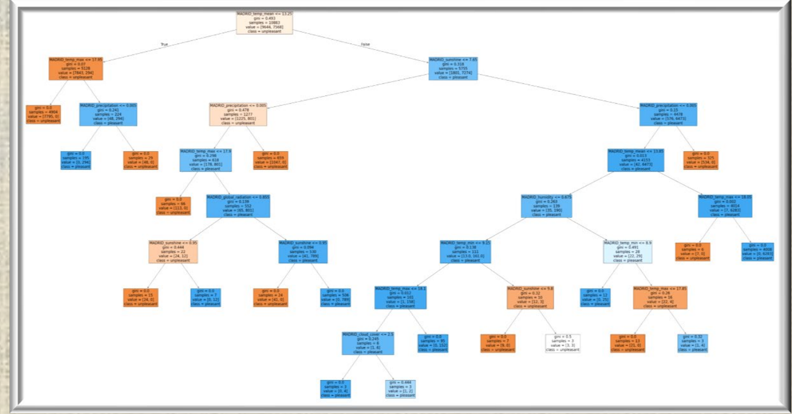
- **Pattern Detection**: Identifies unusual trends by comparing recent data to historical records.

- **Data Needed**:
    - Historical weather data (temperature, humidity, precipitation, wind).
    - Recent/current weather observations for comparison.

What features are important for predicting pleasant days, storms, tornadoes, etc, Does it change over time?



Optimize to find which features are most relevant for the prediction or analysis



Using CNNs for Regional Weather Prediction

How? CNNs:

- **Pattern Recognition:** Analyze satellite and radar images to identify geographic features (mountains, lakes, urban heat islands.

- **Location Mapping:** The network can associate specific areas with historical weather data to predict localized weather patterns.

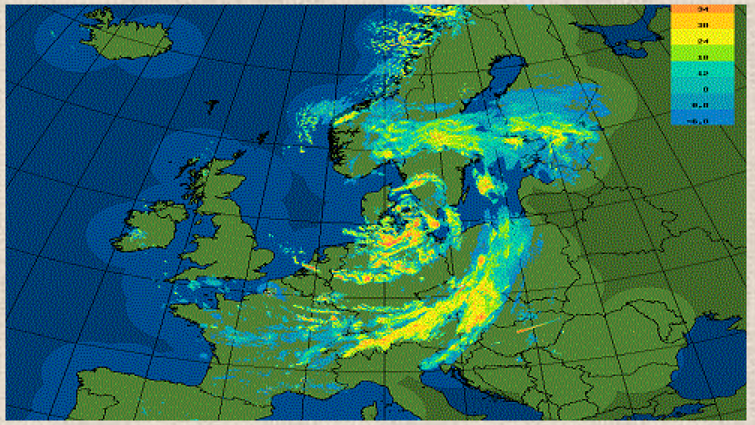
- **Data Needed**:
    - Satellite and radar images.
    - Geographic maps (elevation, coastlines, urban areas).
    - Historical weather data for training.

The CNN might learn:

- Storms that pass over lakes may increase in intensity due to higher humidity.

- Mountain ranges might cause storms to lose strength as they cross, resulting in less rainfall on the opposite side.

- Urban areas might experience stronger heatwaves due to the urban heat island effect.



Deep Learning Image Sensors for Air Quality

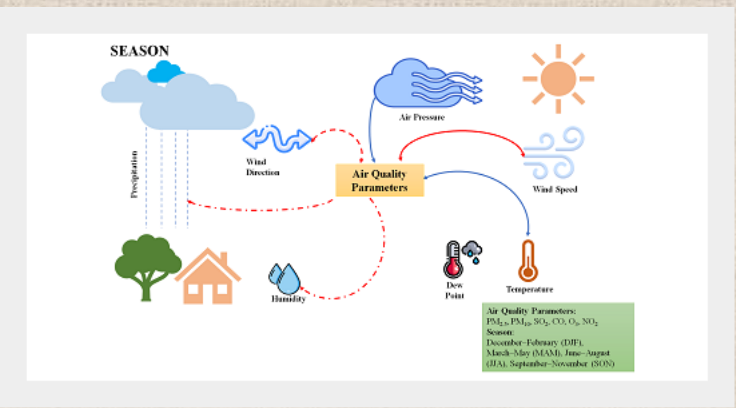
Camera Based Weather Stations & Mobile Drone Stations:

- Increase coverage and lower cost with stations that only include a camera.
- Make use of existing web, surveillance, and traffic cameras.

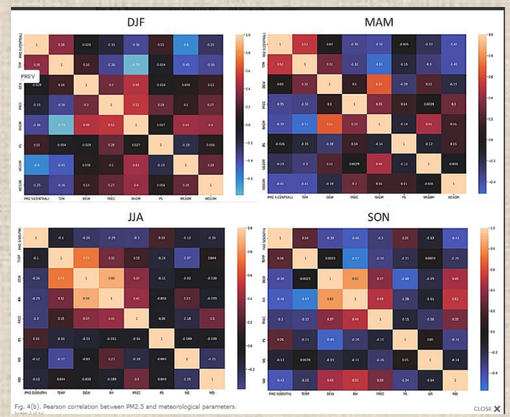
Why?

- Visibility is influenced by factors like *emissions, air pollutants, sunlight, humidity, temperature, and time*.
- Affects how clearly objects can be seen through the atmosphere.
- Train model on images with known humidity, temperature, and air particulates

A photo is really a layered array of data that we can mine if we have the right tools



Istiana, T., Kurniawan, B., Soekirno, S., Nahas, A., Wihono, A., Nuryanto, D.E., Adi, S.P., Hakim, M.L. (2023). Causality Analysis of Air Quality and Meteorological Parameters for PM2.5 Characteristics Determination: Evidence from Jakarta. Aerosol Air Qual. Res. 23, 230014. https://doi.org/10.4209/aaqr.230014



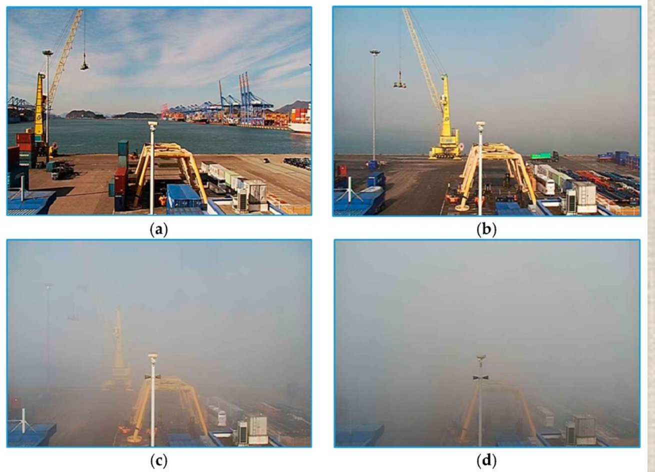
Pearson correlation between $PM_{2.5}$ and meteorological parameters.
*PM2.5 refers to particulate matter that is 2.5 micrometers or smaller in diameter & is an important metric for air quality

Deep Learning Image Sensors for Air Quality

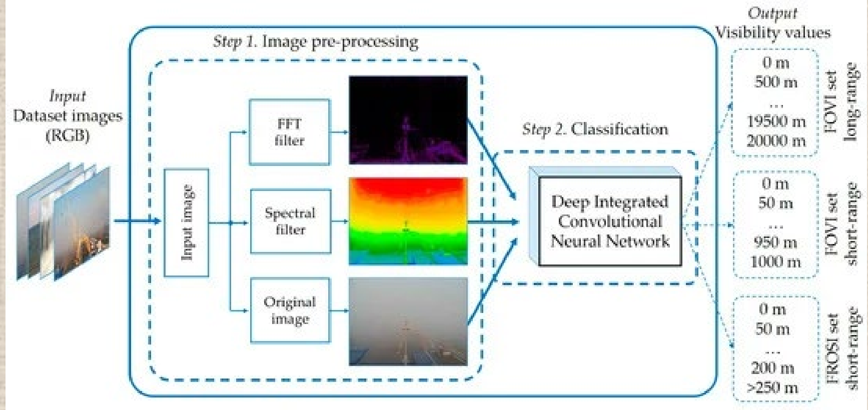
Proof of Concept:
VisNet- deep integrated CNNs:

- Designed to estimate visibility distances from camera images.
- Three parallel streams of CNNs, analyzing images to determine visibility under different foggy weather conditions.

- **Relevance to Air Quality**:
- **Visibility** is influenced by factors like emissions, air pollutants, sunlight, humidity, temperature, and time.
- It is a phenomenon that affects how clearly objects can be seen through the atmosphere.



MDPI and ACS Style
Palvanov, A.; Cho, Y.I. VisNet: Deep Convolutional Neural Networks for Forecasting Atmospheric Visibility. Sensors 2019, 19, 1343. https://doi.org/10.3390/s19061343

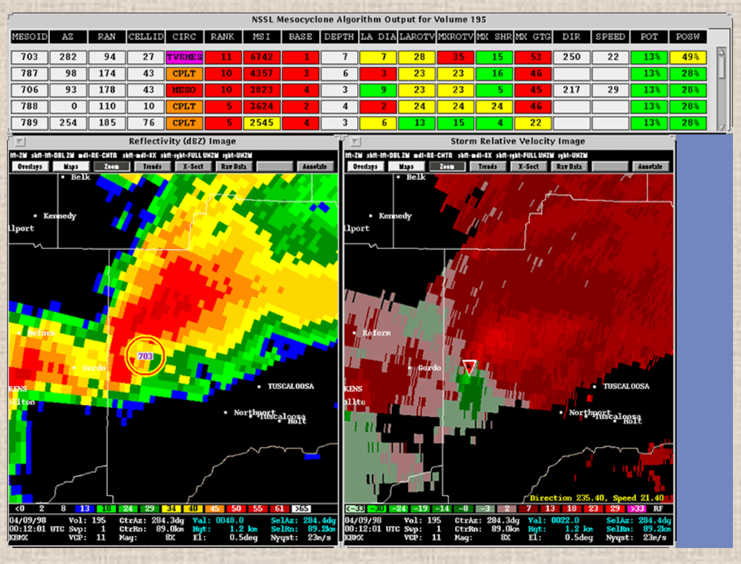


Generative Adversarial Networks (GANs) for Tornado Prediction

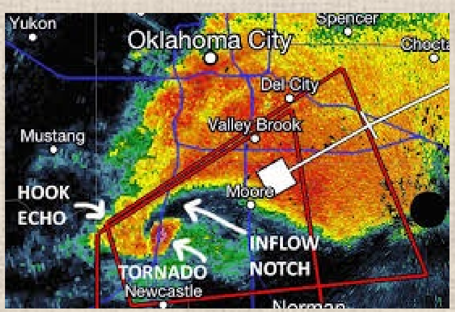
Why GANs for Tornado Prediction?

- **Pattern Discovery**: Can detect complex patterns in radar data that are too subtle for humans.
- **Storm Cell Prediction**: After learning from real data, GANs can predict if a storm cell might generate a tornado.
- **Benefits**:
  - Greater amount of warning time
  - Models will help understand safe vs risky locations and structures

Mesocyclone Detection Algorithm Display



Appearance of tornadoes on radar



Generative Adversarial Networks (GANs) for Tornado Prediction

Tornado Analysis with GANs

- **Learning from Radar Data:**
  - GANs analyze large datasets of radar images, learning the features of tornado-producing storms.
  - The Discriminator gets increasingly accurate at distinguishing radar images of storms that produce tornadoes & those that don't.
- **Predicting Tornado Formation:**
  - Based on learned patterns, the GAN can analyze a new radar image and predict whether the storm will generate a tornado.
- Lincoln Laboratory researchers who curated the dataset, called TorNet, have now released it open source.

Data Needed for GAN-based Tornado Prediction

- **Radar Images**: High-resolution storm cell images, both tornado-producing and non-tornado-producing.
- **Labeled Data**: Annotations indicating whether storms generated tornadoes, including intensity and other patterns.
- **Historical Weather Data**: Temperature, wind speed, humidity, and pressure during storm events.
- **Geographic Information**: Terrain features (mountains, plains) and urban/rural details.
- **Simulated Data (Optional)**: Synthetic storm data to improve rare tornado event predictions.
- **Satellite Imagery (Optional):** Cloud formation and storm development imagery to enhance feature extraction.

Wrapping Up…

Temperature emerged as the key factor in distinguishing pleasant from unpleasant days.

With temperatures rising globally, understanding how it interacts with other weather factors is crucial for accurate forecasting.

Train a model using photos to assess air quality based on visibility and explore how it correlates with weather metrics such as temperature and precipitation.

The stations with the most importance to the model seem to be located in south centrally.

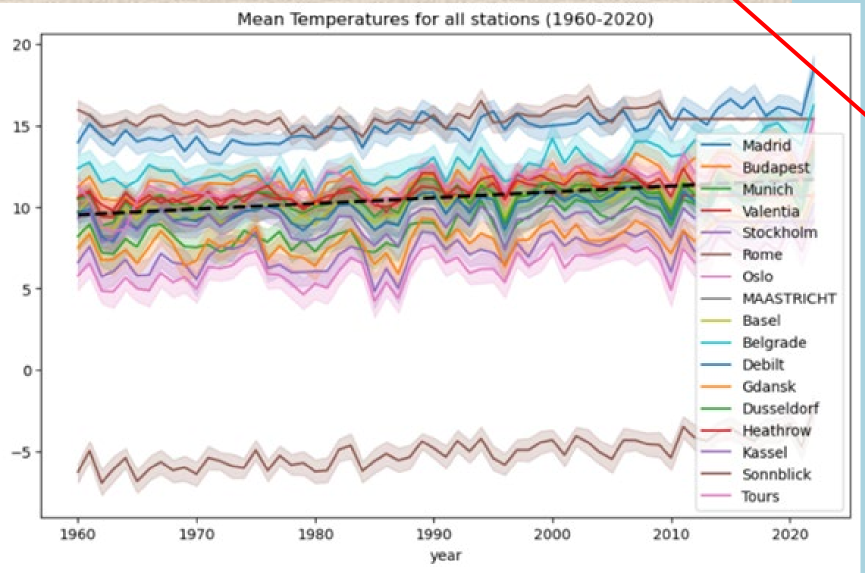
After identifying unusual and extreme weather patterns, examining regional differences and the influence of geographical features can reveal which, if any, have a significant impact.

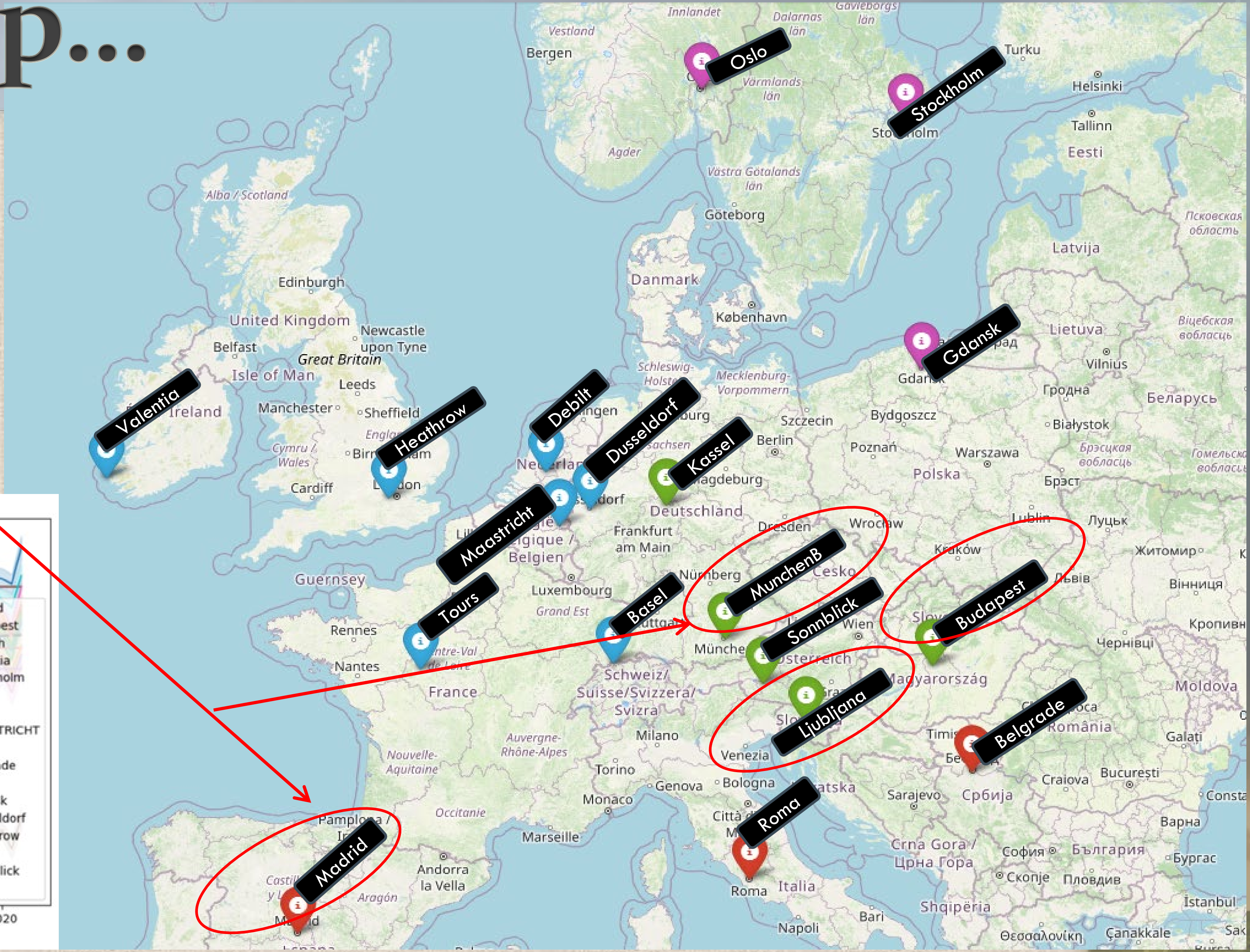
Build and train GAN model w/ radar data from historical confirmed tornadoes & storms.

Understanding how twisters form will enhance the ability to predict tornado dynamics, providing earlier warnings and identifying high-risk areas.

Wrapping Up…

- The stations with the most importance to the model seem to be located in south centrally.
- Roma wasn’t in the model because of the lack of data for it





Things to Think About

What makes weather prediction a good candidate for tackling with machine learning?

How can we avoid human bias entering the analysis process?

Are there other large weather datasets that could complement this analysis?

Thank you for you time!

Tiffany Kinney
tiffkinney@gmail.com